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RATING SOUND INSULATION IN TERMS OF SPEECH INTELLIGIBILITY

PACS: 43.55.Hy

Park, Hyeon Ku, Bradley, John S. and Gover, Bradford N.
National Research Council, Montreal Rd. Ottawa Canada, K1A 0R6

ABSTRACT

This paper describes new listening tests to evaluate various airborne sound insulation ratings in terms of the intelligibility of transmitted speech. In these experiments listeners heard test sentences played, at a fixed level, through 20 different simulated walls in the presence of a constant low-level ambient noise. The results show that the ISO *Weighted Sound Reduction Index* (R_w) and ASTM *Sound Transmission Class* (STC) ratings were not good predictors of the intelligibility of the transmitted speech sounds. However, measures intended to be indicators of the intelligibility of speech, such as the *Articulation Index* (AI) and the *Speech Intelligibility Index* (SII) were more strongly related with the mean intelligibility scores. Other measures based on arithmetic averaging of transmission loss values and/or limiting frequencies to those important for speech, led to better predictors of sound insulation ratings. Two more successful approaches were an arithmetic average of transmission loss values over speech frequencies and a new speech spectrum weighting for the ISO R_w procedure.

INTRODUCTION

Airborne sound insulation ratings can be evaluated in terms of their correlation with various subjective ratings of sound insulation. This paper considers sound insulation ratings in terms of the intelligibility of the transmitted speech because speech is a common type of disturbing sound and because speech intelligibility tests can provide accurate subjective ratings.

Airborne sound insulation is rated in terms of the ISO *Weighted Sound Reduction Index* (R_w) or the ASTM *Sound Transmission Class* (STC). Previously, Vian et al. [1] related subjective ratings of sound insulation to frequency limited (125 Hz – 4k Hz) A-weighted level differences. Tachibana et al. [2] found judgements of the loudness of transmitted sounds to be predicted by a simple arithmetic average transmission loss over frequency. Recent research has shown the intelligibility of speech from meeting rooms to be well related to frequency-weighted signal-to-noise ratios [3], suggesting possible new wall transmission loss ratings.

EXPERIMENTAL PROCEDURES

The listening tests were carried out in a sound isolated and acoustically dead test space. Subjects heard speech sounds, modified to include the transmission characteristics of 20 different walls, and presented from loudspeakers in front of them (See Figure 1). At the same

time, noise with a –5 dB/octave spectrum shape and an overall level of 35 dBA was played from loudspeakers above the subject.

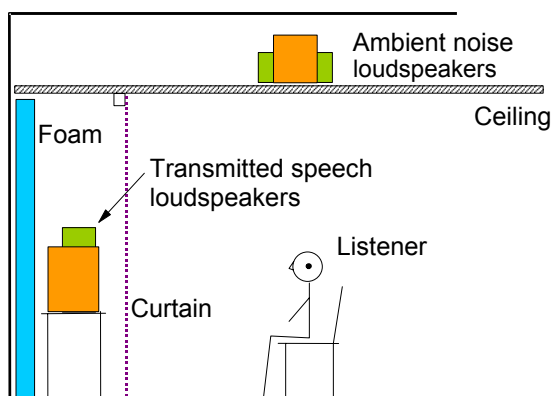


Figure 1. Schematic of cross-section through Room Acoustics Test Space showing the location of the listener and the loudspeakers used to generate the test sounds.

The characteristics of the 20 simulated walls were chosen to represent a range of STC values evenly distributed from STC 34 to 58 (R_w 33 to 56). They include a variety of construction types and transmission characteristics as illustrated in Figure 2.

The speech tests used the Harvard sentences [4]. These are phonetically balanced English sentences with content that is of low predictability, which is important to minimize the effects of guessing. The sentences were all recorded by the same clear-speaking male talker. The speech source level and the ambient noise levels were held constant throughout the tests. Only the sound transmission characteristics of the simulated walls were varied.

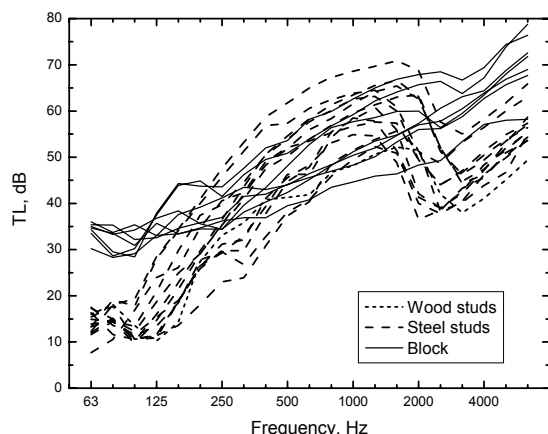


Figure 2. Sound Transmission loss versus 1/3-octave band frequency for the 20 walls simulated in the listening tests, where those containing wood studs, steel studs and concrete blocks are separately identified.

Subjects heard 5 different sentences for each of the 20 different simulated walls. The speech intelligibility scores of the 75 combinations of 5 sentences and 15 subjects per wall were averaged and plotted versus various sound insulation ratings.

Boltzmann equations were fitted to the data for each sound insulation measure and the related R^2 values compared. The Boltzmann equation is given by the following,

$$y = \frac{A_1 - A_2}{1 + e^{(x-x_0)/dx}} + A_2$$

where, A_1 = y-value for $x = -\infty$ (0% or 100%)
 A_2 = y-value for $x = +\infty$ (0% or 100%)
 x_0 = x-value of mean y-value, that is the x-value when $y=50\%$ in our case
 dx = relates to the slope of the mid-part of the regression line.

STANDARD SOUND INSULATION MEASURES

Figure 3 compares the variation of STC values and mean speech intelligibility scores and their standard errors versus wall number with the walls ordered in terms of increasing STC value. Although the even distribution of STC values is evident, the mean intelligibility scores do not closely follow the same trend.

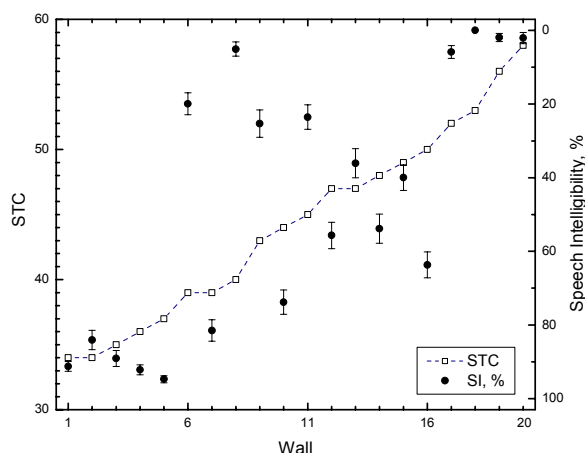


Figure 3. Mean speech intelligibility scores with error bars indicating ± 1 standard deviation and Sound Transmission Class (STC) versus wall number.

Figure 4 Plots mean intelligibility score versus R_w ratings for the 20 simulated walls. The intelligibility scores are not well related to the R_w values The plot versus STC values produced similar results.

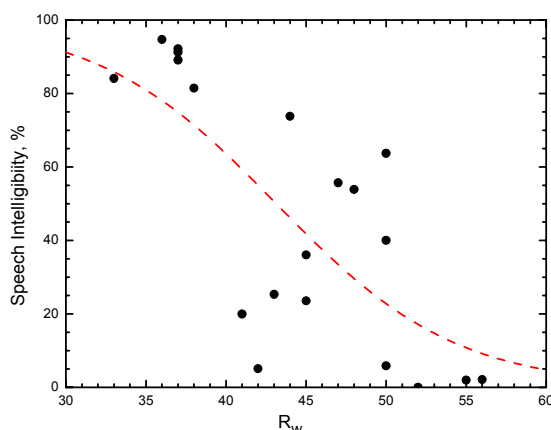


Figure 4. Speech intelligibility scores versus R_w ratings of the simulated walls. ($R^2 = 0.542$).

Further analyses were carried out to explore the details of the STC procedure. First, the '8 dB rule' for the maximum allowed deviation included in the STC procedure was investigated. New STC values were calculated in which the 8 dB rule was varied from a 0 dB to a 16 dB rule. Figure 5 shows the correlation coefficients of these new measures with the mean intelligibility scores. These results suggest that increasing the maximum allowed deficiency to 14 dB or more would increase the correlations with intelligibility scores, as would eliminating the 8 dB rule.

The merits of the maximum allowed combined deficiency of 32 dB, included in both the ASTM and the ISO standards, was similarly tested. The correlation coefficients with mean intelligibility scores in Figure 6 show that the maximum combined deficiency should be 30 dB or more. These results support the use of a 32 dB maximum allowed combined deviation.

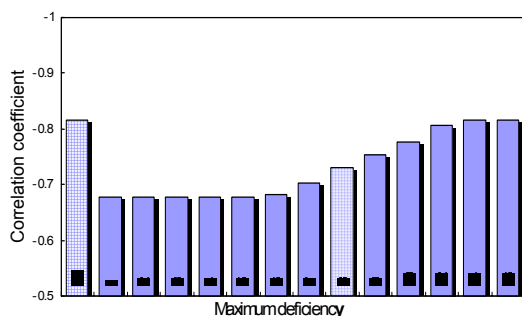


Figure 5. Correlation coefficients between mean speech intelligibility scores and a modified STC value for which the maximum acceptable deficiency was varied from 1 to 16 dB, including a no maximum deficiency case in first bar at the left.

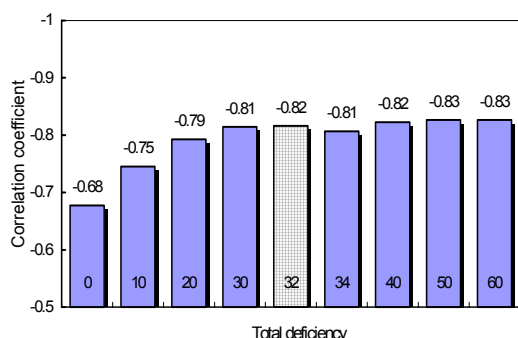


Figure 6. Correlation coefficients between speech intelligibility scores and modified STC values for which the total acceptable deficiency was varied from 0 to 60 dB. (Calculations did not include an 8-dB rule).

The ISO Standard for airborne sound insulation measurement includes the option of various spectrum correction terms. One is useful for rating sound insulation against outdoor road traffic noise. Another is intended to produce results similar to an A-weighted reduction to a pink noise source. Neither of these spectrum correction terms was helpful in providing better relationships with the speech intelligibility scores. However a new spectrum weighting term was developed that did very successfully improve the relationship with intelligibility scores. The spectrum weighting ($C_{400-2.5k}$) simply included equally all frequencies from 400 to 2.5k Hz and rejected all

other frequencies. Figure 7 shows this new measure to be very strongly related to speech intelligibility scores.

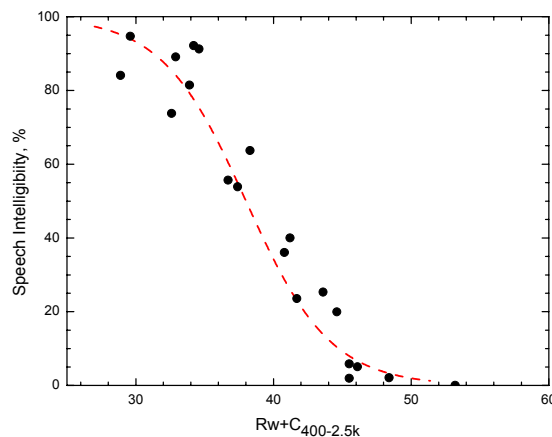


Figure 7. Speech intelligibility scores versus R_w ratings with an added $C_{400-2.5k}$ -type spectrum weighting correction for the simulated walls. ($R^2 = 0.957$)

SPEECH INTELLIGIBILITY TYPE MEASURES

A number of measures were evaluated that are derived from or related to measures of speech intelligibility. The Articulation Class (AC) is a single number attenuation rating derived from the Articulation Index (AI) and includes the same frequency weightings. Figure 8 shows intelligibility scores to be quite well related to the AC ratings of the walls. The Articulation Index values were similarly well related because in these experiments speech source levels and ambient noise levels were held constant.

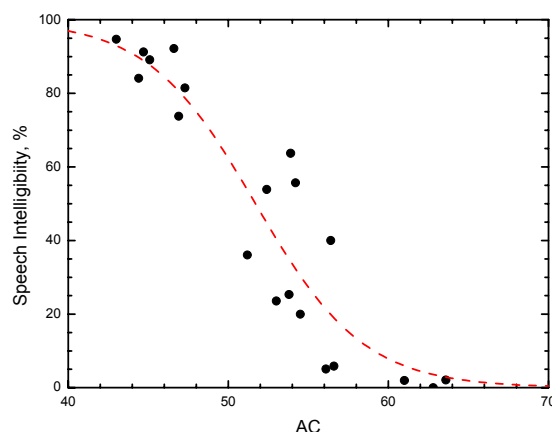


Figure 8. Speech intelligibility scores versus the AC ratings of the simulated walls. ($R^2 = 0.856$)

Although many have suggested using A-weighted speech-noise level differences at the listener's position ($SNR(A)$) as a simple sound insulation rating, the results in Figure 9 suggest that this is not a good indicator of the expected intelligibility of the transmitted speech.

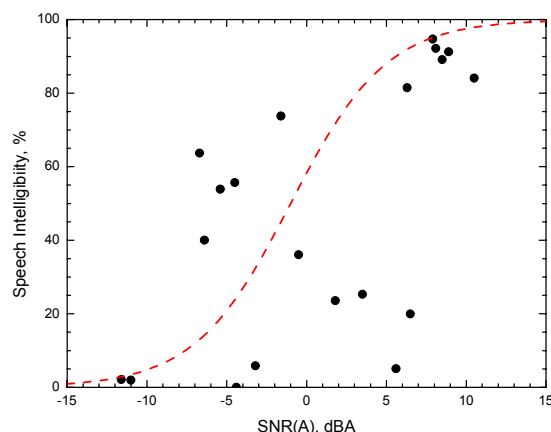


Figure 9. Mean speech intelligibility scores versus $SNR(A)$ ratings of the test conditions. ($R^2 = 0.259$)

Recent work [3] on the speech security of meeting rooms showed that frequency-weighted signal-to-noise ratios of the transmitted speech and ambient noise levels were good predictors of speech security. In the previous work a weighted signal-to-noise ratio (SNR_{sli22}) with the same

frequency weightings as the Speech Intelligibility Index (SII) was found to best predict the intelligibility of the transmitted speech. Figure 10 shows that these SNR_{SII22} values were equally well related to the intelligibility scores in the present study.

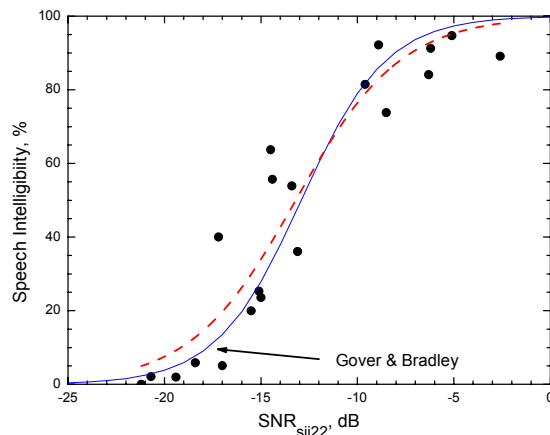


Figure 10. Speech intelligibility scores versus SNR_{SII22} ratings of the test conditions. Solid line labelled 'Gover & Bradley' is from previous experiment [3]. ($R^2=0.913$)

The AI, SII, AC, SNR_{SII22} and several other related measures are all based on arithmetic averages of decibel values over a range of frequencies. Other variations of this concept were also considered including uniform-weighted averages of transmission loss values over various frequency ranges. The most successful was an arithmetic average of transmission loss values from 200 to 2.5k Hz referred to as AA(200-2.5k). In Figure 11 mean intelligibility scores are plotted versus AA(200-2.5k) values. Of all the measures evaluated, this measure produced the strongest relationship with intelligibility scores.

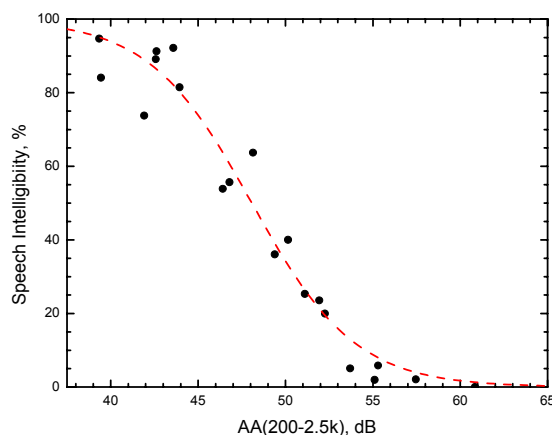


Figure 11. Speech intelligibility scores versus AA(200-2.5k) ratings of the simulated walls. ($R^2=0.959$).

DISCUSSION

Table 1 summarizes the results presented in this paper and others from the complete study [5]. The standard sound insulation ratings STC and R_w were not strongly related to intelligibility scores. However, removing the 8 dB rule from the STC rating and limiting the included frequency range improved the R^2 values.

Measures in which decibel values over a range of frequencies are arithmetically averaged, were all generally quite successful. These included: AA(160-5k), AA(200-2.5k), AI, SII, SNR_{AI} , SNR_{SII22} , SNR_{uni32} , and AC values. However, when measures included *energy averaging* of values at various frequencies, results tended to be less successful (e.g. dBavg(160-5k)).

The arithmetic average transmission loss measure AA(200-2.5k) and the R_w measure with the new spectrum weighting term $C_{400-2.5k}$ provide very good relationships with mean speech intelligibility scores and are considerable improvements over existing standard measures. The new spectrum weighting is also appealing because it adds to an existing standardized approach. However, these new ratings must now be tested in terms of responses to other types of sounds with different acoustical characteristics than speech.

Symbol	F ₁ , Hz	F ₂ , Hz	R ² Boltzmann	Description
STC	125	4k	0.510	Standard STC
STC _{no8}	125	4k	0.661	STC without an 8 dB rule
STC _{no8}	200	4k	0.846	Modified STC without an 8 dB rule
STC _{no8}	200	2.5k	0.922	Modified STC without an 8 dB rule
R _w	100	3.15k	0.542	Standard R _w
R _w	200	3.15k	0.842	Modified R _w
R _w	200	2.5k	0.922	Modified R _w
R _w +C	100	3.15k	0.359	R _w with C-type spectrum correction
R _w +C _{tr}	100	3.15k	0.205	R _w with C _{tr} -type spectrum correction
R _w +C _{spp}	100	3.15k	0.864	R _w with C _{spp} -type spectrum correction
R _w +C _{400-2.5k}	100	3.15k	0.951	R _w with C _{sps} -type spectrum correction
R _w +C _{Opt}	100	3.15k	0.9657	R _w using simplified speech shape rating contour
R _{wOpt}	100	3.15k	0.951	R _w using simplified speech shape rating contour
AA(160-5k)	160	5k	0.853	Arithmetic average transmission loss
AA _{opt} (200-2.5k)	200	2.5k	0.959	Arithmetic average transmission loss
AI	200	5k	0.864	Articulation Index
SII	160	8k	0.899	Speech Intelligibility Index
SNR _{ai}	200	5k	0.896	AI-weighted signal-to-noise ratio
SNR _{sii22}	160	8k	0.913	SII-weighted signal-to-noise ratio
SNR _{uni32}	160	5k	0.853	Uniform weighted signal-to-noise ratio
SNR(A)	50	8k	0.259	A-weighted speech – noise level difference
AC	200	5k	0.856	Articulation Class
dBavg(160-5k)	160	5k	0.446	Energy average of transmission loss values
STA	100	5k	0.361	A-weighted transmission loss
Loudness	25	12.5k	0.305	Zwicker loudness, Sones

Table 1. Summary of R² values for Boltzmann equation fits to plots of speech intelligibility scores versus various sound insulation ratings as well as lower (F₁) and upper (F₂) included frequency.

Acknowledgements

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